# SMART DISPLAY MODULE SPECIFICATION

| 4.3 Inch Smart Display with TOUCH | |
|---|---|
| **Model:** | PLCM-UES31S043H800V480C-U |
| **MCU：** | ESP32-S31-WROOM-3 |
| **PCB：** | UES31S043H800V480C-U |
| **Display：** | 4.3" IPS, 800 × 480, ST7262, RGB666/888, GT911 CTP |
| **Version:** | V1.1 |
| **Date:** | 2026-10-15 |

VIEWE TECHNOLOGY CO.，LTD

**Customer Confirmation**

| Approved by | Notes |
|---|---|
| | |

# REVISION HISTORY

| Revision | Date | Contents of Revision Change | Remark |
|---|---|---|---|
| V1.0 | 20260903 | First release | |
| V1.1 | 20261015 | Add power supply description, purchase link , wiki link, and product images. | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

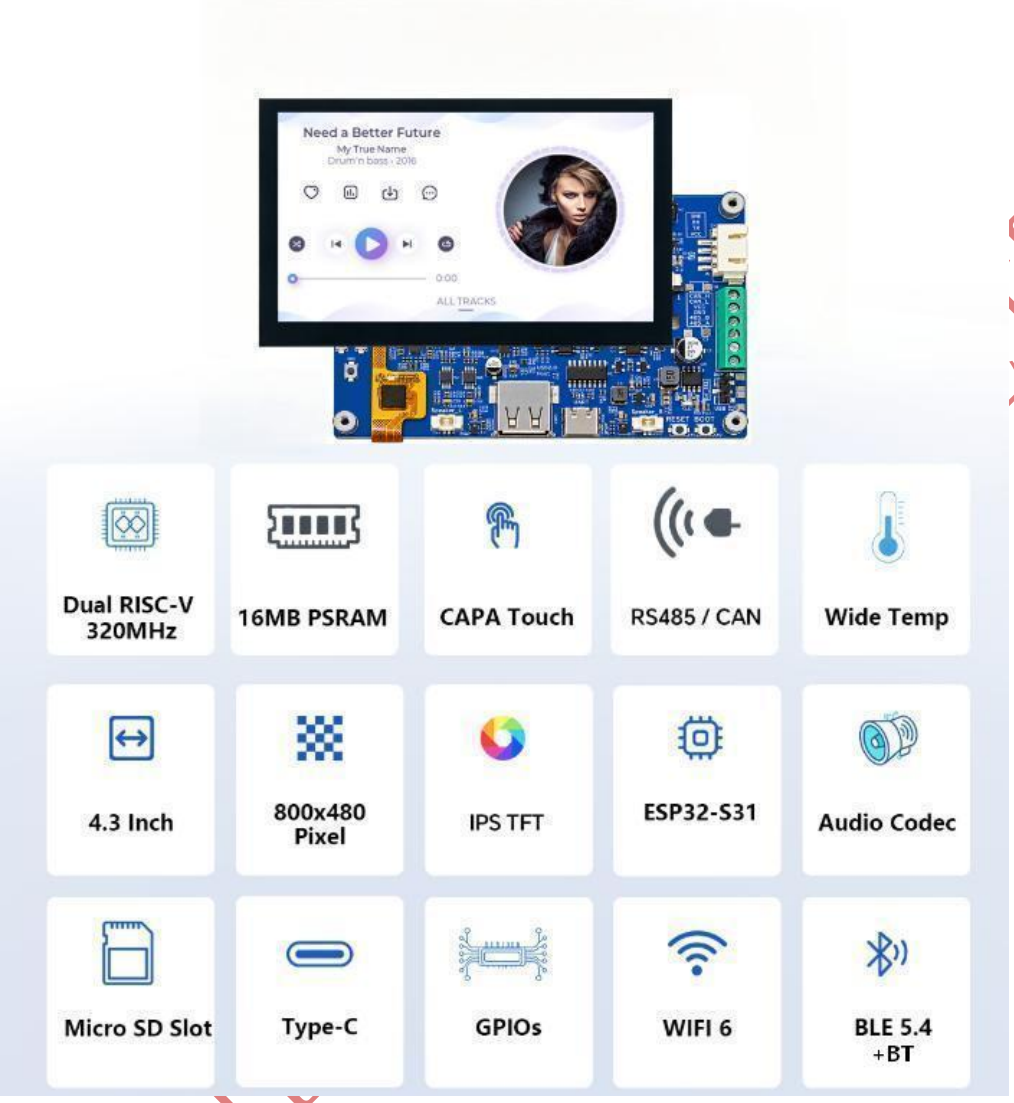
VIEWE TECHNOLOGY CO., LTD

# TABLE of CONTENTS

**1. INTRODUCTION** ..... **4**
1.1 Product Features ..... 5
1.2 Applications ..... 6
1.3 Product Naming ..... 6
**2. PRODUCT INFORMATION** ..... **7**
2.1 Interface Description ..... 7
2.2 GPIO Definition ..... 10
2.3 Display Information ..... 11
2.4 LCD RGB timing (verified) ..... 11
2.5 Audio Interface ..... 11
2.6 H1 header (2 × 10, 2.54 mm), top to bottom ..... 12
2.7 USB 2.0 Interface ..... 12
2.8 UART / RS485 / CAN ..... 13
2.9 Storage, Keys, LED ..... 14
2.10 Voltage & Current ..... 14
2.11 Reliability Test ..... 14
**3. FUNCTIONAL BLOCK DIAGRAM** ..... **15**
**4. MECHANICAL DRAWING** ..... **16**
**5. DETAILED INFORMATION (CODES, DRAWINGS...)** ..... **17**

VIEWE TECHNOLOGY CO., LTD

# 1. Introduction




PLCM-UES31S043H800V480C-U is a high-performance smart display module designed by VIEWE. It is based on the Espressif ESP32-S31-WROOM-3 module and a 4.3-inch RGB capacitive touch panel (800 × 480). The LCD and touch devices are the same as those used on UEDX80480043E-WB-B (ST7262 driver IC and GT911 CTP). The main board is a new design around ESP32-S31: a dual-core 32-bit RISC-V MCU running at up to 320 MHz, with Wi-Fi 6, Bluetooth 5.4 (LE) plus Bluetooth Classic, USB 2.0 High-Speed OTG, stereo audio, SDIO 3.0 storage, and RS485 / CAN industrial interfaces.Supports USB-C 5V input or DCIN 7 – 24V wide voltage input.

**Notice:** ESP32-S31 is a preview device. ESP-IDF builds must use the --preview option (for example: idf.py --preview set-target esp32s31). Module ratings follow the Espressif ESP32-S31-WROOM-3 datasheet (pre-release). New designs that need USB 2.0 High-Speed, Wi-Fi 6, or on-board codec audio should use this S31 board rather than the ESP32-S3。

## 1.1 Product Features

**CPU:**

1. Processor
   1) ESP32-S31 RISC-V 32-bit dual-core processor, main frequency up to 320 MHz, plus a ULP-RISC-V coprocessor.
   2) 2.4 GHz Wi-Fi 6 (IEEE 802.11b/g/n/ax, HT20/40, up to 150 Mbps), Bluetooth 5.4 LE (including LE Audio), Bluetooth Classic (BR/EDR), and IEEE 802.15.4 (Zigbee / Thread).
   3) On-module PCB antenna.
   4) Hardware security: Secure Boot, Flash/PSRAM encryption, cryptographic acceleration, and TEE. See the chip datasheet for details.

2. Memory
   1) On-chip: 320 KB ROM, 512 KB SRAM, 32 KB low-power SRAM.
   2) Typical module configuration on this product: 16 MB Quad SPI Flash + 16 MB Octal SPI PSRAM (ESP32-S31-WROOM-3-N16R16V).
   3) Flash and PSRAM can be accessed in parallel.

3. Peripheral Interfaces
   1) USB Type-C port for 5 V power, firmware download, and serial debug (CH340C bridged to UART0).
   2) USB Type-A port connected to the dedicated USB 2.0 High-Speed OTG PHY (USB_DP / USB_DM, not GPIOs). Host 5 V is sourced through TPS2051C (~500 mA). VBUS-EN is hard-wired high and is not MCU-controlled.
   3) On-board MicroSD slot, 4-bit SDMMC / SDIO 3.0.
   4) ES8389 stereo codec, dual NS4150B speaker amplifiers, dual analog microphones, and L/R speaker connectors.
   5) RS485 (SIT3088E, hardware automatic direction) and CAN (SIT1050T + on-chip TWAI) on a 6-pin terminal.Supports DCIN 7 – 24V wide voltage input.
   6) 2 × 10 pin 2.54 mm header (H1) bringing out UART0/UART1, I2C, RS485/CAN GPIOs, 3.3 V / 5 V / GND.
   7) Four ADC keys, RESET / BOOT, WS2812B RGB LED, and a 3 kHz passive buzzer（Not attached by default）.

   For more information on ESP32-S31-WROOM-3, please refer to the Espressif S31 datasheet .

**Display**

1) Size: 4.3 Inch
2) Resolution: 800 *480
3) Pixel Arrangement：RGB Vertical Stripe
4) Interface Mode：40PIN RGB 24bits
5) Driver IC： ST7262E43-G4

VIEWE TECHNOLOGY CO., LTD

6) Touch IC: GT911
7) Brightness: 400 cd/m$^2$
8) Touch: CTP
9) Verified timing: PCLK 18 MHz, negative polarity; HSYNC 1 / 40 / 20; VSYNC 1 / 10 / 5.

More information about Display can be found here:Display Specification.pdf

**Other**

1) Operation Temperature: −20 ~ 70 ° C (limited by the LCD).
2) Storage Temperature: −30 ~ 80 ° C.
3) MCU module rating: −40 ~ 85 ° C.

## 1.2 Applications

With rich connectivity and powerful processing capabilities, the UEDX80480043E-WB-B is an ideal choice for IoT devices in the following areas:

- Smart Home Control Panels
- Industrial Automation HMI
- Smart Appliances
- Consumer Electronics
- Wireless Data Loggers
- Touch Screen Interfaces
- Educational Learning Platforms

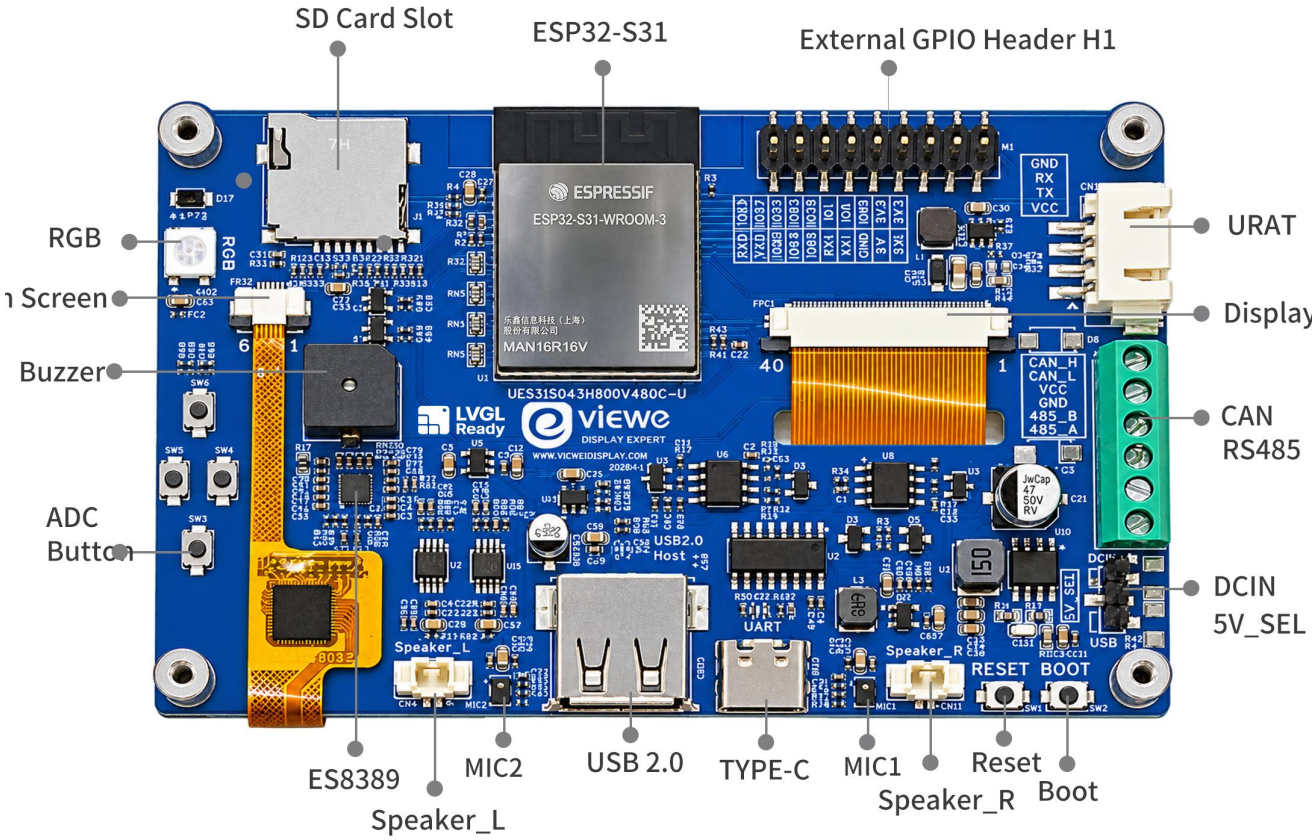
## 1.3 Product Naming

| Field | Code | Meaning |
|---|---|---|
| Form | PLCM | PCB+LCM(Display) |
| Brand / series | UE | VIEWE smart display module |
| MCU | S31 | ESP32-S31-WROOM-3 |
| Size | S043 | 4.3 inch |
| Resolution | H800V480 | 800 × 480 |
| Touch | C | Capacitive (GT911) |
| Bus | U | UART/RS485/I2C/CAN/USB 2.0 High-Speed OTG |

# 2. Product information

## 2.1 Interface Description

This module achieves efficient communication with the main control unit via UART, RS485 and CAN buses, and supports USB 2.0 High-Speed OTG functions to meet diverse data interaction requirements. The capacitive touch screen adopts the GT911 solution to deliver smooth and responsive touch experience. The 4.3-inch screen features an 800×480 high-definition resolution, ensuring clear and delicate display content. Integrated with the ESP32-S31-WROOM-3, it balances high-performance processing and low-power operation. Suitable for various industrial and commercial application scenarios, it provides stable and reliable visual interaction support for devices.



**1、Main Control Module:**
ESP32-S31-WROOM-3 Dual-core RISC-V, up to 320 MHz, 16 MB Flash + 16 MB PSRAM, PCB antenna.

**2、Display Interface:**
40-pin RGB output. The panel is 24-bit capable; this board wires RGB666 (R2–R7 / G2–G7 / B2–B7). See the Display Interface table.

**3、SD Card Slot：**
SPI interface connected for external storage expansion, Please refer to Section 2.9 below for detailed information.

**4、Touch interface：**
I2C (SDA = GPIO0, SCL = GPIO1) to GT911. INT and RST are present on the 6-pin FPC; verified examples leave them unused (NC).

**5、USB Type-C interface：**
5 V DC input, programming, and serial debug through CH340C

**6、USB Type-A:**
USB 2.0 High-Speed OTG PHY (USB_DP / USB_DM). Host 5 V via TPS2051C. VBUS-EN is pulled high by hardware.Please refer to Section 2.7 below for detailed information.

**7、UART helper (4-pin):**
GND / RX / TX / VCC for an external serial adapter.The 4-pin pitch of 2.5 mm

**8、RGB-LED (WS2812B)：**
one WS2812B, DIN = GPIO37, powered from 5 V with a level-shift network.Please refer to Section 2.9 below for detailed information.

**9、Boot Buttons：**
BOOT button (GPIO61) for firmware download mode.

**10、Reset Buttons：**
RESET button (CHIP-EN) for resetting the board.

**11、External GPIO header H1:**
2 × 10 pins, 2.54 mm. Please refer to Section 2.6 below for detailed information.

**12、Industrial terminal CN3 (CAN/RS485):**
CAN_H, CAN_L, VCC, GND, 485_B, 485_A， Please refer to Section 2.8 below for detailed information.

**13、Audio:**
Dual- channel speaker interface Speaker L / R (CN2 / CN1) with a 2-pin pitch of 1.25 mm, and on - board dual microphones MIC1 / MIC2 deliver high-quality audio via ES8389 and NS4150B，Please refer to Section 2.5 below for detailed information.

**14、ADC Button：**
ADC key array SW3–SW6 on GPIO42. Please refer to Section 2.9 below for detailed information.

**15、Buzzer:**
BEEP_EN = GPIO46, high = on, 3 kHz passive type.Not attached by default

**16、DCIN / 5V_SEL:**
Select the power supply using the jumper cap. Supports USB-C 5V input or DCIN 7 – 24V wide voltage input.

VIEWE TECHNOLOGY CO., LTD

**Display Interface:**

| Pin No. | Symbol | I/O | Description |
|---|---|---|---|
| **1** | **LEDK** | P | Power supply for backlight cathode |
| **2** | **LEDA** | P | Power supply for backlight anode |
| **3** | **GND** | P | Power ground |
| **4** | **VDD** | P | Logic supply, 3.3 V |
| **5-12** | **R0-R7** | I | Red data. This board uses R2–R7 (GPIO2–7). R0/R1 = NC |
| **13-20** | **G0-G7** | I | Green data. This board uses G2–G7 (GPIO8–13). G0/G1 = NC |
| **21-28** | **B0-B7** | I | Blue data. This board uses B2–B7 (GPIO14–19). B0/B1 = NC |
| **29** | **GND** | P | Power ground |
| **30** | **CLK** | I | Pixel clock, negative polarity (GPIO40) |
| **31** | **DISP** | I | Standby mode. Normally pulled high |
| **32** | **HSYNC** | I | Horizontal sync, negative polarity (GPIO44) |
| **33** | **VSYNC** | I | Vertical sync, negative polarity (GPIO45) |
| **34** | **DEN** | I | Data enable. Display access is enabled when DE is “H” (GPIO43) |
| **35** | **NC** | I | Dummy |
| **36** | **GND** | P | Power ground |
| **37** | **XR** | - | Dummy |
| **38** | **YD** | - | Dummy |
| **39** | **XL** | - | Dummy |
| **40** | **YU** | - | Dummy |

I：Input；O：Output；P：Power

**TP Interface:**

| Pin No. | Symbol | I/O | Description |
|---|---|---|---|
| **1** | **RST** | P | Touch reset on FPC; not used |
| **2** | **3.3V** | P | 3.3 V logic supply |
| **3** | **GND** | P | Power ground |
| **4** | **INT** | I | Touch interrupt on FPC;TP INT = GPIO38 |
| **5** | **SDA** | I | SDA = GPIO0, shared with ES8389 |
| **6** | **SCL** | P | SCL = GPIO1, 400 kHz, shared with ES8389 |

VIEWE TECHNOLOGY CO., LTD

## 2.2 GPIO Definition

| GPIO Number | Current Usage | Function Description |
|---|---|---|
| CHIP-EN | Reset | RESET button (SW1) |
| GPIO0 | I2C SDA | GT911 + ES8389. Pulled high at reset (normal boot). Not the BOOT key; also on H1 |
| GPIO1 | I2C SCL | GT911 + ES8389; also on H1 |
| GPIO2–7 | LCD-R2–R7 | LCD red data, high 6 bits |
| GPIO8–13 | LCD-G2–G7 | LCD green data, high 6 bits |
| GPIO14–19 | LCD-B2–B7 | LCD blue data, high 6 bits |
| GPIO20–23 | SD-D0–D3 | SDMMC 4-bit data |
| GPIO24 | SD-CLK | SDMMC clock |
| GPIO25 | SD-CMD | SDMMC command |
| USB_DP / USB_DM | USB 2.0 HS PHY | Dedicated module pins. Do not configure as GPIO |
| GPIO33 | UART1 TX | Verified as UART1 TX. Also a USB Serial/JTAG pad |
| GPIO34 | UART1 RX | Verified as UART1 RX |
| GPIO35 | RS485 TX | SIT3088E DI; S8050 auto-drives DE/~RE; also on H1 |
| GPIO36 | RS485 RX | SIT3088E RO; also on H1 |
| GPIO37 | WS2812 DIN | On-board XL-5050RGBC-WS2812B |
| GPIO39 | LCD-BL-EN | Backlight enable, active high |
| GPIO40 | LCD-PCLK | Pixel clock, 18 MHz, negative polarity |
| GPIO42 | ADC KEY | SW3–SW6 resistor ladder, ADC1, ~0–2 V usable range |
| GPIO43 | LCD-DE | Data enable |
| GPIO44 | LCD-HS | Horizontal sync (GPIO number, not USB_DM) |
| GPIO45 | LCD-VS | Vertical sync (GPIO number, not USB_DP) |
| GPIO46 | BEEP-EN | Buzzer, active high; 3 kHz PWM recommended |
| GPIO47 | PA-EN | NS4150B amplifier enable, active high |
| GPIO48 | I2S MCLK | Present on schematic; firmware examples leave NC (codec no_mclk) |
| GPIO49 | I2S BCK | ES8389 bit clock |
| GPIO50 | I2S WS | ES8389 word select / LRCK |
| GPIO51 | I2S DOUT | MCU → ES8389 DAC |
| GPIO52 | I2S DIN | ES8389 ADC → MCU |
| GPIO53 | CAN TX | SIT1050T TXD |
| GPIO54 | CAN RX | SIT1050T RXD |
| GPIO55–57 | H1 GPIO | Broken out on H1 |
| TX0（IO58）/ RX0(IO59) | UART0 | CH340C download / log; also on H1 |
| GPIO60 | SD-CTRL | SD 3.3 V power switch, active low |
| GPIO61 | BOOT | BOOT button (SW2); also on H1 |

SD I2C Reset
RS485 UART1 USB2.0
Buzzer LCD RGB
CAN I2S PA-EN
GPIO UART0 Boot

## 2.3 Display Information

| Item | Specification | Unit | Remark |
|---|---|---|---|
| **Pixel Driving element** | IPS TFT | - | - |
| **Screen Size** | 4.3 | Inch | Diagonal |
| **Resolution** | 800(W)*3(RGB)*480(H) | Dots | - |
| **Interface** | RGB 24bits | - | 40PIN |
| **Module Power Consumption** | 1.04 | Watt | Typ. |
| **Active Area** | 95.04(W)*53.86(H) | mm | - |
| **Pixel pitch (W*H)** | 0.1188(W)*0.1122(H) | mm | - |
| **Module Size (W*H*D)** | 105.52(W)*67.17(H)*2.8(D) | mm | Tolerance:±0.2 |
| **Luminance** | 400 | cd/m$^2$ | Typ. |
| **Viewing Direction** | All | O'clock | - |
| **Display Color** | 16.2M | Colors | 24bits |

## 2.4 LCD RGB timing (verified)

| Parameter | Value |
|---|---|
| **PCLK** | 18 MHz, pclk_active_neg = true |
| **Horizontal resolution** | 800 |
| **Vertical resolution** | 480 |
| **HSYNC pulse / back porch / front porch** | 1 / 40 / 20 |
| **VSYNC pulse / back porch / front porch** | 1 / 10 / 5 |
| **Framebuffer** | RGB888 in PSRAM, one frame buffer |

## 2.5 Audio Interface

The audio codec is Everest ES8389 (I2C 7-bit address 0x20). I2S0 is used in full duplex. 48 kHz / 16-bit / stereo playback and record have been verified. MCLK is not required (codec configured with no_mclk). Dual NS4150B Class-D amplifiers drive the left and right speakers. PA_EN = GPIO47. MIC1 and MIC2 are on-board analog microphones.

| Signal | GPIO / Device | Description |
|---|---|---|
| I2C SDA / SCL | GPIO0 / GPIO1 | Shared with GT911, 400 kHz |
| I2S BCK | GPIO49 | Bit clock |
| I2S WS | GPIO50 | Word select / LRCK |
| I2S DOUT | GPIO51 | DAC playback |
| I2S DIN | GPIO52 | ADC record |
| I2S MCLK | NC in firmware | Optional on schematic |
| PA_EN | GPIO47 | High = amplifier on |
| Speaker | CN1 / CN2 | NS4150B differential output, typ. 4 Ω / 3 W class |

## 2.6 H1 header (2 × 10, 2.54 mm), top to bottom

| Number | Left | Number | Right |
|---|---|---|---|
| 1 | 3V3 | 2 | 5V |
| 3 | 3V3 | 4 | 5V |
| 5 | GND | 6 | GND |
| 7 | GPIO0 (SDA) | 8 | TX1 (GPIO33) |
| 9 | GPIO1 (SCL) | 10 | RX1 (GPIO34) |
| 11 | GPIO35 (485_TX) | 12 | GPIO36 (485_RX) |
| 13 | GPIO53 (CAN_TX) | 14 | GPIO54 (CAN_RX) |
| 15 | GPIO55 | 16 | GPIO56 |
| 17 | GPIO57 | 18 | TX0 |
| 19 | GPIO61 (BOOT) | 20 | RX0 |

GPIO35/36 and GPIO53/54 on H1 are in parallel with the on-board RS485 and CAN transceivers. Do not drive both the header and the terminal at the same time. GPIO0/1 already have I2C pull-ups.

## 2.7 USB 2.0 Interface

ESP32-S31 integrates a USB 2.0 High-Speed OTG PHY on dedicated USB_DP / USB_DM pins (package pins 44 / 45 of the chip). These are not GPIOs and need no GPIO-matrix setup. The board provides two physical connectors:

| Port | Bridge / PHY | Role | Notes |
|---|---|---|---|
| Type-C | CH340C → UART0 | Download / log / 5 V in | Hold BOOT (GPIO61), then tap RESET |
| Type-A | USB 2.0 HS PHY | OTG Host or Device | TPS2051C sources 5 V; VBUS-EN hard-wired high |

Verified applications: (1) TinyUSB HID mouse mapped from the touch panel — class-compliant on Windows and Linux, no custom driver; (2) USB extended display — host-side IDD / vendor driver required, JPEG decode to the RGB panel. Type-A defaults to supplying 5 V as a USB Host. When the same port is used as a Device attached to a PC, avoid back-feeding VBUS from both ends.

## 2.8 UART / RS485 / CAN

| Interface | Controller | MCU pins | External | Verified setting |
|---|---|---|---|---|
| UART0 | CH340C | TX0 / RX0 | Type-C / H1 | Download and monitor |
| UART1 | GPIO matrix | TX=33, RX=34 | H1 TX1 / RX1 | 115200 8N1 |
| RS485 | SIT3088E + UART2 | TX=35, RX=36 | CN3 485_A / 485_B / GND | 115200 8N1, auto DE |
| CAN | SIT1050T + TWAI | TX=53, RX=54 | CN3 CAN_H / CAN_L / GND | 500 kbit/s, Classic CAN |

Note：Supports USB-C 5V input or DCIN 7 – 24V wide voltage input.

**CN3 terminal (follow the PCB silkscreen)**

| Silk | Function |
|---|---|
| CAN_H | CAN high |
| CAN_L | CAN low |
| VCC | Supports USB 5V power supply or DCIN 7-24V wide voltage power supply (selected via jumper cap) |
| GND | Common ground |
| 485_B | RS485 B |
| 485_A | RS485 A |

RS485: DE/~RE is switched automatically from 485_TX through an S8050. Treat the port as a normal UART in software; do not configure RTS. 120 Ω termination and A-pull-up / B-pull-down are on board. CAN: 120 Ω termination on board; CAN_GND is tied to system ground through 0 Ω. Do not connect a USB-TTL adapter to A/B or CAN_H/L.

## 2.9 Storage, Keys, LED

**Micro SD**

SDMMC 4-bit: CLK = GPIO24, CMD = GPIO25, D0–D3 = GPIO20–23, SD_CTRL = GPIO60 (active low). Examples mount FAT at SDMMC_FREQ_HIGHSPEED. Assert GPIO60 low before initializing the host.

**ADC keys (measured on the real board)**

| Key | Typical voltage | Software window | Remark |
|---|---|---|---|
| Idle | ≈ 3.3 V (pull-up) | Saturated / idle | S31 ADC 0 dB ≈ 0–2 V; idle saturates |
| SW3 | ≈ 0.38 V | 100–600 mV | GPIO42 |
| SW4 | ≈ 0.82 V | 600–1080 mV | GPIO42 |
| SW5 | ≈ 1.34 V | 1080–1605 mV | GPIO42 |
| SW6 | ≈ 1.87 V | 1605–2200 mV | GPIO42 |

Two keys cannot be detected at once (inherent to the divider). Silkscreen order on some drawings does not match the measured voltages; use the table above.

## 2.10 Voltage & Current

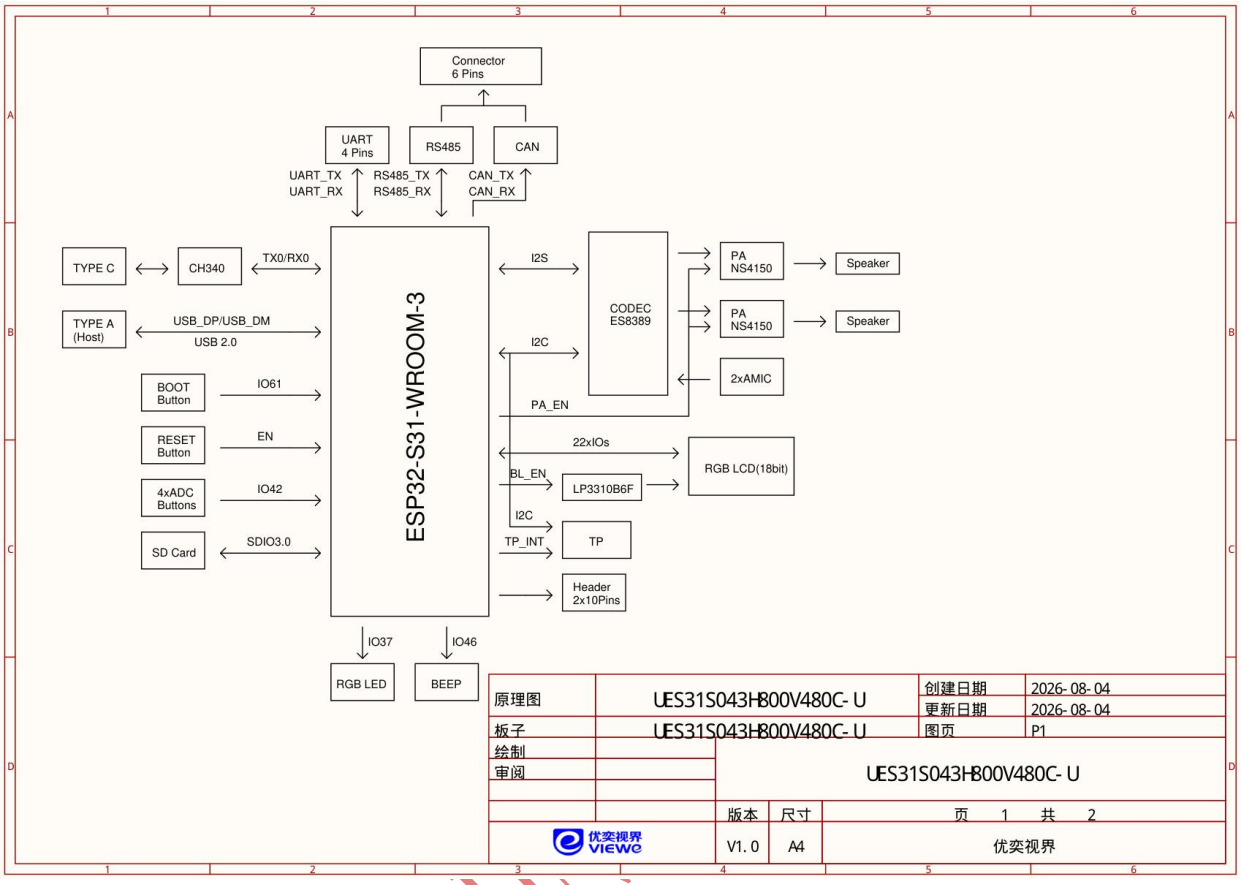
| Item | Conditions | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| Power Voltage | DC | - | 5.0 | - | V |
| | DC | 7 | - | 24 | V |
| Operation Current | VCC= +24V, Maximum backlight current | - | 100 | - | mA |
| | VCC= +12V, Maximum backlight current | - | 200 | - | mA |
| | VCC= +5V, Maximum backlight current | - | 400 | - | mA |
| | VCC= +5V,backlight off | - | 150 | - | mA |
| Recommended power supply:5V 1A DC | | | | | |

Note: 5V powers the USB-C port, or 12V and 24V power the DCIN ports.Select the power supply using the jumper cap.

## 2.11 Reliability Test

| Item | Conditions | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| Working Temperature | 60%RH at 5V voltage | -20 | 25 | 70 | C |
| Storage Temperature | --- | -30 | 25 | 85 | C |
| Working Humidity | 25°C | 10% | 60% | 90% | RH |
| ESD | --- | Contact: ±4KV<br>Air: ±8KV | | | KV |

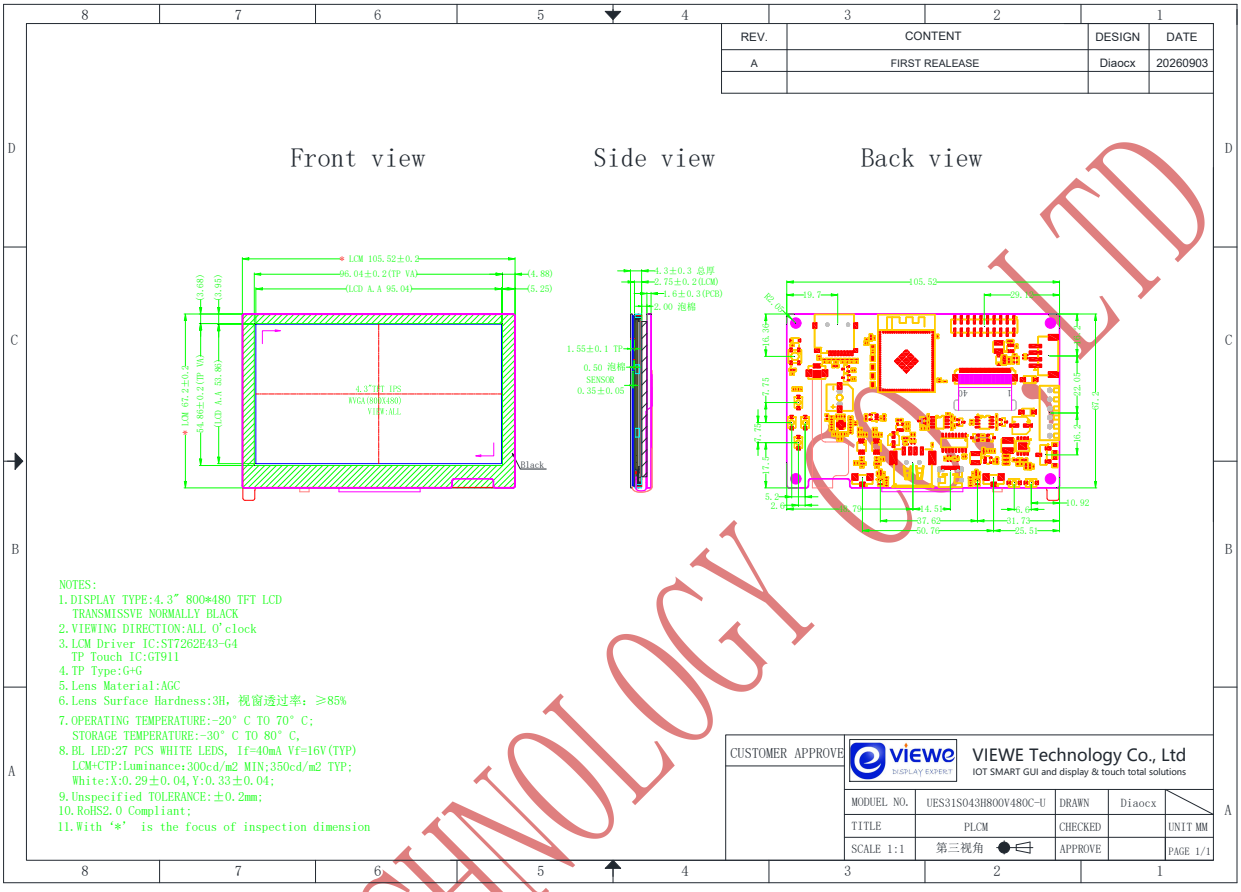
# 3. Functional Block Diagram




**Note:** ESP32-S31 includes a 2.4 GHz radio that supports Wi-Fi 6, Bluetooth 5.4 (LE), Bluetooth Classic, and 802.15.4. Because they share the same RF front-end, Wi-Fi and Bluetooth cannot transmit or receive simultaneously; the radio switches between protocols as needed. USB 2.0 High-Speed is an independent PHY and does not share that front-end.

VIEWE TECH

# 4. MECHANICAL DRAWING




VIEWE TECHNOLOGY CO.,LTD

# 5. Detailed information (codes, drawings...)

**Github：** https://github.com/VIEWESMART/PLCM-UES31S043H800V480C-U

**Gitee:** https://gitee.com/VIEWESMART/PLCM-UES31S043H800V480C-U

**Wiki：** https://displaywiki.com/products/esp32-s3-4.3inch-800x480-touch-uart-display-hmi

**Official Store：**

https://viewedisplay.com/product/4-3-inch-800x480-esp32-s31-wifi6-touch-smart-hmi-uart-display

**Prompt:** The materials are being developed and enriched. Please stay tuned.

VIEWE TECHNOLOGY CO.,LTD